

Internship Project Report

Improvement of the Position Measurement System of a Milling Machine Using a Digital Readout (DRO) System

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Abstract

This project focuses on the study, design, and proposed implementation of a Digital Readout (DRO) position-measurement system for a manual universal milling machine. In the target machine, axis positions are determined manually using mechanical graduations and vernier scales. This approach depends on operator reading and can introduce reading errors, increase setup time, and reduce positioning accuracy. The project therefore investigates the existing milling machine, studies another company milling machine already equipped with a DRO as a reference, evaluates linear measurement technologies, selects a suitable DRO architecture, and develops the mechanical installation concept and three-dimensional CAD model.

The proposed system uses a three-axis digital readout for the X, Y, and Z axes together with linear magnetic measurement components. The work also includes examination of installation constraints, alignment, cable routing, protective arrangements, available DRO functions, market alternatives, and possible supplementary solutions such as an edge finder and touch probe. The report additionally documents a separate finite-element stress analysis of a perforated scrubber filter carried out as an engineering activity during the internship.

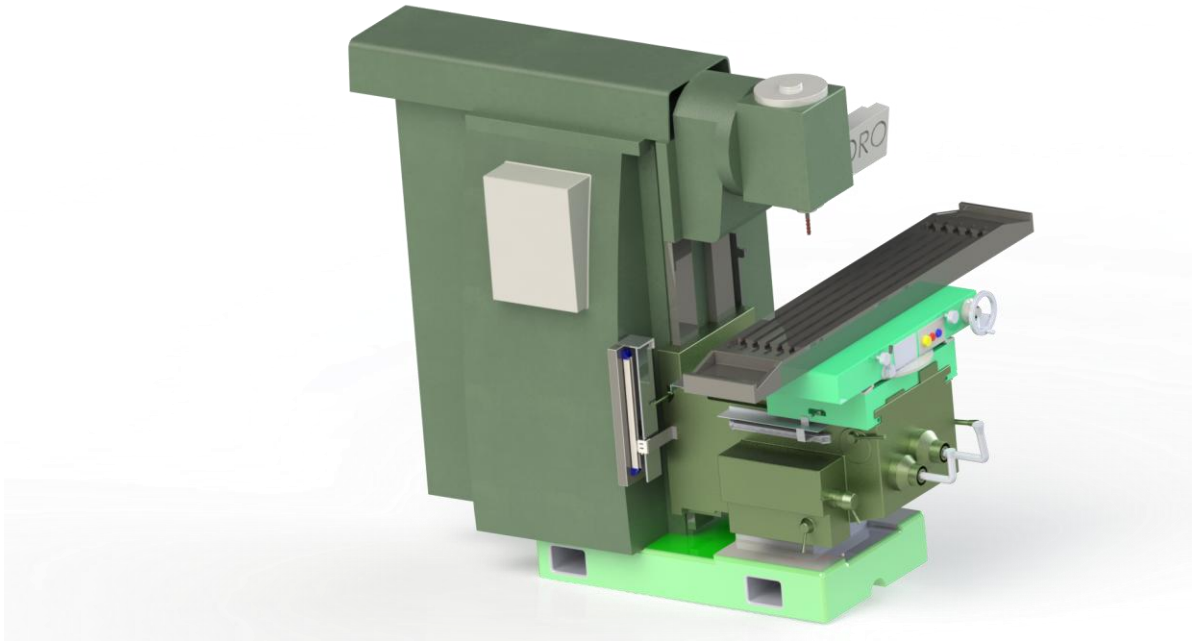
The implementation roadmap includes fabrication, installation, calibration, and performance testing. These stages are presented as planned/conditional activities in the report and should not be interpreted as completed experimental validation unless explicitly documented.

Table of Contents

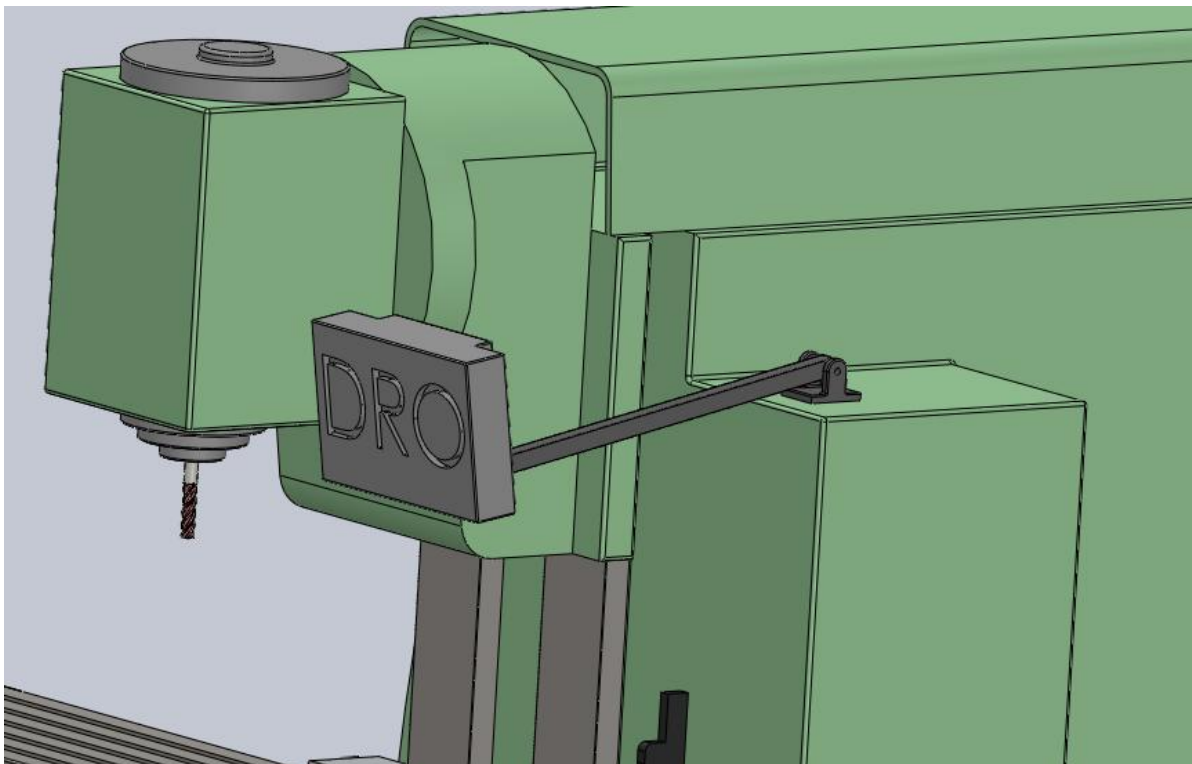
1. Introduction
2. Main Project Objective
3. Project Roadmap
4. Company Introduction
5. Why Use a Digital Readout (DRO)?
6. Proposed Project Schedule
7. Preliminary Study of Milling Machine 2 (Without DRO)
8. Study of Milling Machine 1 (Equipped with DRO)
9. Study of the Existing DRO System
10. Magnetic Linear Measurement System
11. Market Comparison of Position Measurement Systems
12. Alternative Optimization Solutions
13. CAD Renderings
14. Finite Element Analysis of a Perforated Scrubber Filter

13.1 Selected CAD Renderings

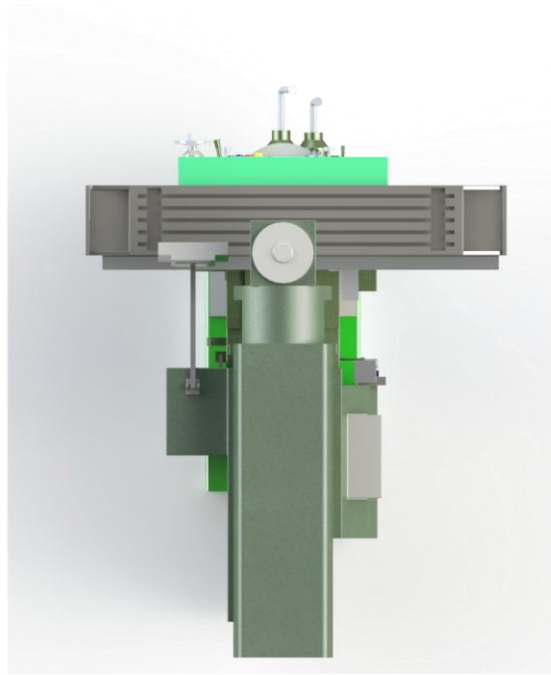
The following renderings are included to visually document the proposed DRO installation, the overall machine assembly, and representative mounting arrangements for the linear measurement components.



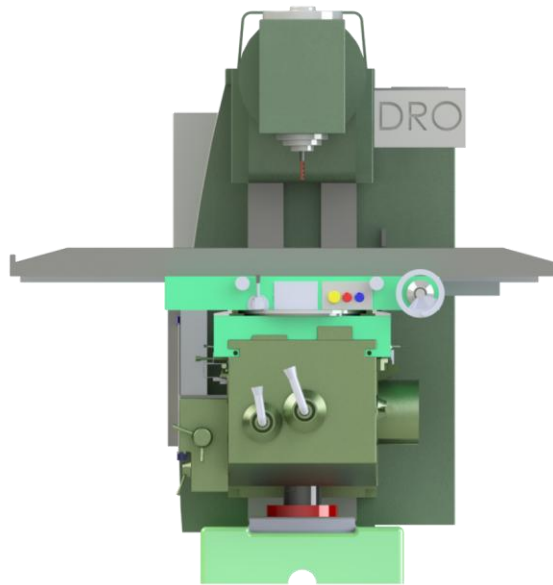
Overall CAD rendering of the milling machine and proposed DRO installation.



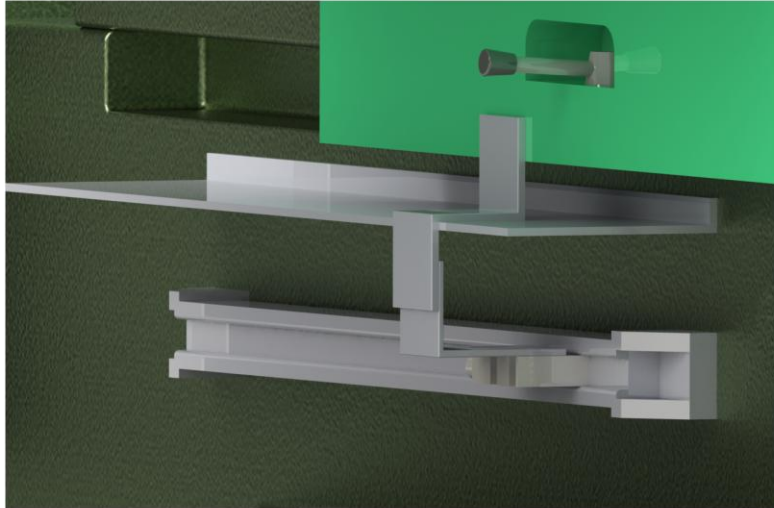
Close-up CAD rendering of the DRO display and its mounting location.



CAD view of the milling head and DRO installation arrangement.



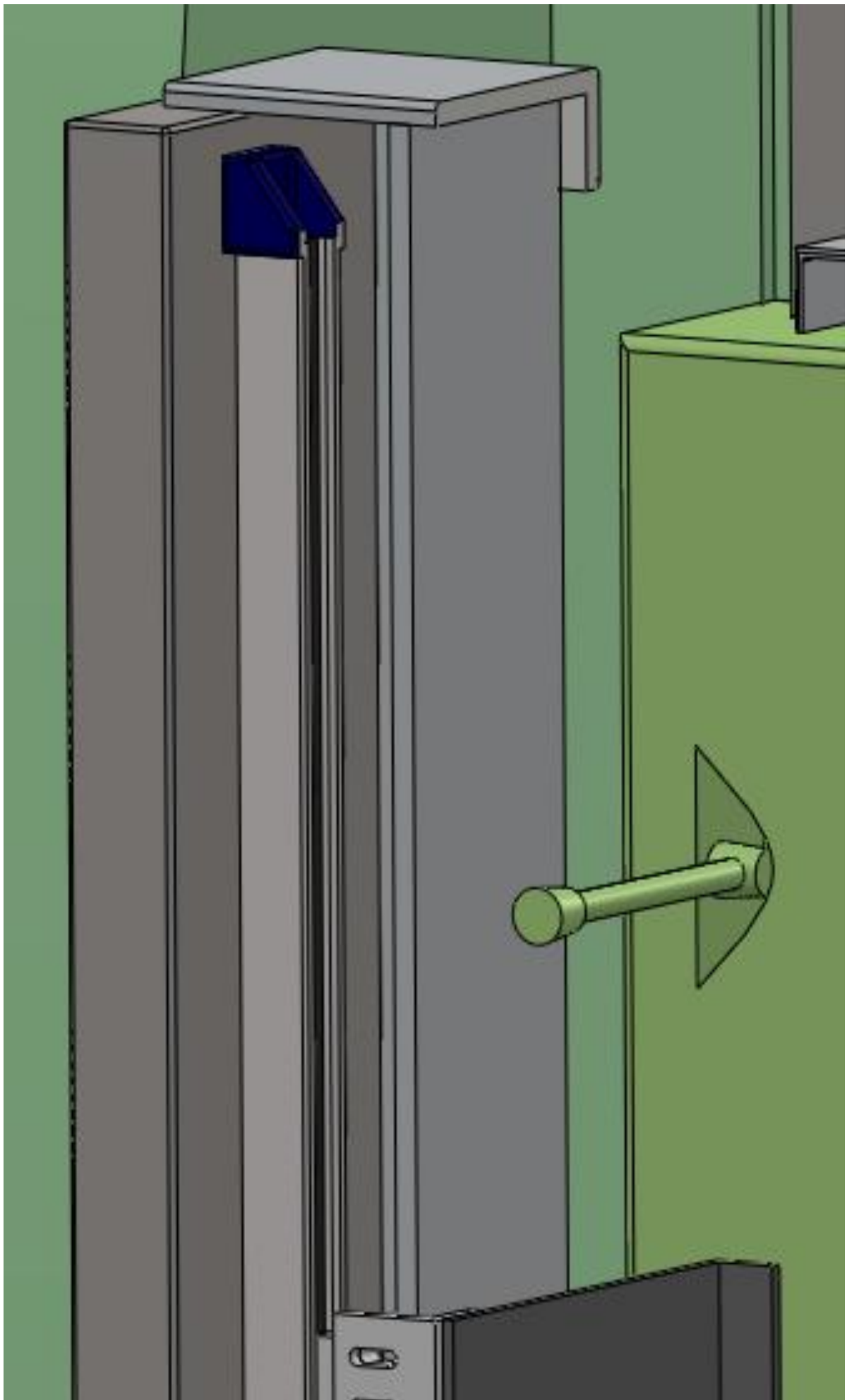
Front-oriented CAD rendering showing the proposed installation concept.



CAD rendering of the linear measurement arrangement on the machine.



Detailed CAD view of a linear scale/read-head installation.



Detailed CAD view of the vertical-axis linear measurement installation.



CAD view of the table/axis measurement installation.

15. Conclusion

1. Introduction

Manual milling machines are widely used in machining operations, and accurate positioning of the machine axes has a direct effect on the dimensional quality of manufactured parts. In the target milling machine, the position of the axes is currently determined manually using mechanical graduations. Although this method is conventional and useful, it depends strongly on the operator and may lead to reading errors, longer setup times, and reduced positioning accuracy.

A second milling machine in the company is equipped with a Digital Readout system. This machine provides a practical reference for studying the installation and operation of a DRO and for developing an upgrade concept for the target machine. Alternative measurement and optimization methods are also considered so that the final approach can be evaluated from technical, economic, installation, and maintenance perspectives.

2. Main Project Objective

The main objective is to develop an engineering solution for upgrading the second milling machine by adding a digital position display system similar to the system used on the reference machine. During table or axis movement, the actual measured displacement can be displayed continuously on the DRO, reducing the operator's dependence on reading mechanical vernier graduations.

- Improve the practical accuracy of axis-position measurement.
- Reduce errors associated with manual reading of mechanical graduations.
- Reduce setup time during machining operations.
- Make axis positioning easier for the operator.
- Improve repeatability and consistency of machining setup.

- Upgrade the measurement capability without converting the complete machine to CNC.

Before final selection, alternative solutions are considered based on required accuracy, cost, installation complexity, environmental conditions, and maintainability.

3. Project Roadmap

3.1 Study and Identification of the Milling Machine

Study the overall machine structure, X/Y/Z axis movement, transmission mechanisms, table movement, current position-measurement method, technical specifications, and present condition.

3.2 Study of the Existing Measurement System

Evaluate mechanical graduations/verniers, the reading method, achievable resolution, axis backlash, and limitations affecting positioning accuracy.

3.3 Study of the DRO-Equipped Reference Machine

Study the DRO unit, linear scales, read heads, mounting locations, display position, mechanical connections, and cable routing.

3.4 Dimensional Survey and Installation Conditions

Measure the X/Y/Z travel ranges and identify suitable spaces for linear scales, read heads, brackets, cables, and protective elements. Potential interference, chips, oil, and vibration are considered.

3.5 Selection of the System and Required Equipment

Select or specify the DRO, linear measurement components, and accessories according to accuracy, resolution, measuring range, working environment, and available installation space.

3.6 Installation-System and Mechanical-Part Design

Develop the mounting concept for each axis and design required brackets and mechanical components. SolidWorks is used for three-dimensional modeling and engineering drawings.

3.7 Design and Mechanical-Interference Check

Check strength, alignment, interference, unrestricted axis travel, maintenance access, and cable routing; revise the design when necessary.

3.8 Cost Estimation

Estimate the cost of the DRO, linear scales, accessories, brackets, mechanical parts, cabling, and installation where applicable. Compare alternative solutions from both technical and economic viewpoints.

3.9 Three-Dimensional Model and Installation Simulation

Prepare a three-dimensional model of the milling machine and the DRO system, including linear scales, read heads, brackets, display unit, and cable paths. The model is used to visually communicate the installation concept and axis operation.

3.10 Fabrication and Procurement

After design approval, manufacture or procure the required brackets and mechanical components and prepare the DRO equipment for installation.

3.11 DRO Installation

Install the linear scales, read heads, display unit, connections, and cables. Carefully adjust mounting positions and alignment.

3.12 Calibration and System Adjustment

Check the DRO on the X, Y, and Z axes and compare displayed displacement with a reference measurement method.

3.13 Performance Testing

Evaluate measurement error under different axis movements and assess accuracy, usability, and setup time.

3.14 Result Comparison and Improvement Assessment

Compare the post-installation condition with the original measurement method in terms of measurement accuracy, reading error, usability, and setup speed.

3.15 Documentation and Final Presentation

Document equipment specifications, drawings, design work, implementation images, calibration/testing results, and the final comparison.

4. Company Introduction

Andisheh Shomal Machine Manufacturing Co. is an industrial and manufacturing company in Gilan Province, Iran, involved in the design, manufacture, and supply of industrial equipment and products. The company uses technical personnel, workshop equipment, and engineering facilities to manufacture and supply equipment for different industrial sectors, particularly the oil, gas, and petrochemical industries.

A significant part of the company's activities concerns the design and manufacture of industrial filtration equipment. During the internship, the company environment provided practical exposure to machine tools, mechanical manufacturing, CAD modeling, engineering drawings, technical documentation, and engineering analysis.

5. Why Use a Digital Readout (DRO)?

A Digital Readout is a position-measurement and display system used on machine tools. Unlike CNC, a DRO does not control the machine's axes. It measures axis displacement through linear measurement devices and displays the measured position to the operator.

- Digital display of X, Y, and Z positions.
- Reduced dependence on mechanical graduation reading.
- Faster establishment of workpiece coordinates.
- Convenient zeroing and incremental/absolute coordinate functions.
- Linear and segmented error-compensation capabilities.
- Useful geometric and drilling functions.
- Potential reduction in setup time and operator workload.

6. Proposed Project Schedule

The proposed schedule divides the project into machine study, measurement-system assessment, reference-machine investigation, dimensional survey, equipment selection, mechanical design, CAD modeling, procurement/fabrication, installation, calibration, testing, comparison, and documentation. The actual completion of later implementation stages depends on the available internship time and project progress.

7. Preliminary Study of Milling Machine 2 (Without DRO)

7.1 Machine Information

Parameter	Specification
Manufacturer	Arsenal
Model	ΦY 321M / FU 321M
Machine type	Universal milling machine with vertical spindle/head configuration
Serial number	N 8698
Year	1992
Country	Bulgaria
Axes	3 (X, Y, Z)
Table size	1350 × 320 mm
Table rotation	45°
T-slots	5 slots; 63 × 18 mm
X-axis travel	Approximately 1000 mm; automatic travel 980 mm
Y-axis travel	Approximately 500 mm; automatic travel 480 mm
Z-axis travel	Approximately 460 mm; automatic travel 440 mm
Table-to-spindle distance	30–490 mm
Spindle taper	ISO 50
Spindle speeds	18 speeds; 32–1600 rpm
Main motor	7.5 kW
Feed motor	2.2 kW
X/Y feed	10–1000 mm/min
Z feed	4–415 mm/min
Rapid X/Y	2500 mm/min
Rapid Z	1040 mm/min
Machine mass	2780 kg
Existing position measurement	Mechanical graduations / vernier
Existing DRO	None

7.2 Main Machine Components

- Milling head
- Spindle
- Tool holder
- Drawbar
- Milling cutter
- Column
- Knee
- Table
- Saddle
- Base

- T-slots
- X-axis handwheel
- Y-axis handwheel
- Z-axis handwheel
- Lead screw
- Axis locks
- Gibbs
- Workholding system
- Spindle-speed control
- Lubrication system

7.3 Axis Motion and Position Measurement

The machine provides manual axis movement through handwheels as well as mechanical feed capability. The existing position measurement is based on mechanical graduations and vernier reading. This method provides useful operator feedback but introduces dependence on visual reading and mechanical conditions such as backlash.

7.4 Operator Discussion

Operator feedback was considered as part of the investigation, particularly regarding the practical process of positioning the axes, reading the graduations, setting the workpiece, and the limitations encountered during machining setup.

7.5 Documentation

The original report includes photographs and documentation of the target machine, its main components, and the existing measurement arrangement.

8. Study of Milling Machine 1 (Equipped with DRO)

The first milling machine in the company was studied as a reference machine because it already uses a digital position-measurement system. The machine is a JAFO FWF 32J2 universal milling machine manufactured in Poland in 1991.

Parameter	Specification
Manufacturer	JAFO
Model	FWF 32J2
Country	Poland
Year	1991
Table size	1300 × 320 mm
Approximate mass	2800 kg
T-slots	5; 18 mm slot width; 63 mm spacing
X travel	850 mm
Y travel	340 mm
Z travel	500 mm
Spindle speeds	16; 56–1800 rpm
Vertical milling head	Available; approximately 45° rotation
DRO	Installed

The reference machine is used to understand practical DRO installation, the relationship between linear measurement components and the display unit, mounting locations, cable routing, and operator interaction with the system.

9. Study of the Existing DRO System

9.1 DRO Datasheet

The reference DRO is an ATEK Sensor Technologies ADR10-3 three-axis digital readout manufactured in Turkey. It receives incremental encoder signals from the linear measurement system installed on the machine and calculates/displays the instantaneous X, Y, and Z positions.

Parameter	Specification
Manufacturer	ATEK Sensor Technologies
Model	ADR10-3
Axes	3-axis X/Y/Z
Encoder input	Incremental encoder
Signals	A/A, B/B, Z/Z line-driver signals
Connector	D-Sub 9-pin encoder connector
Units	Metric / Inch
Selectable resolution	0.1, 0.2, 0.5, 1, 5, 10, 25 μ m or user-defined
Display	8 digits + sign digit, green display
Display range	-99999.999 to +99999.999 mm
Power supply	85-265 VAC
Maximum power	10 W
Dimensions (H $\times$ W $\times$ T)	202 $\times$ 320 $\times$ 84 mm
Approximate mass	2.7 kg
Housing	Aluminium injection body
Coordinate memory	Up to 1000
Communication	RS-232

9.2 Functions and Capabilities

ABS / INC System

Absolute mode displays position relative to a defined origin; Incremental mode allows a temporary work origin for subsequent measurements.

Metric / Inch Measurement

Allows the operator to select millimetres or inches according to the drawing or machining requirement.

One-Touch Axis Reset

Allows an axis to be quickly reset to zero at a selected position. W is an auxiliary/work coordinate rather than an independent mechanical axis.

Axis Selection

Sx, Sy, and Sz keys select the X, Y, and Z axes for DRO functions. Four-axis configurations may also include Sw.

Circular Drilling (Divisor)

Calculates coordinates of holes distributed around a circle from parameters such as centre, radius/diameter, number of holes, and angle.

Shrink Function

Allows a shrinkage factor to be included in dimensional calculations. It is of limited relevance to ordinary milling operations.

1/2 Function

Quickly calculates half of a distance or coordinate, which is useful for finding workpiece centres.

Trigonometric Calculator

Provides sin, cos, and tan calculations for geometry and angled machining operations.

Linear Error Correction

Applies a correction factor to compensate for systematic linear measurement error along an axis.

Segmented Error Correction

Divides an axis into sections and applies individual correction values. The ATEK ADR10 system supports up to 100 correction points.

Finding the Workpiece Center

Uses measured edge/point positions to determine the workpiece centre. A touch probe can make the process faster and more accurate, but is not mandatory.

Digital (Vibration) Filter

Reduces the influence of measurement-signal noise and vibration on the displayed value. It does not eliminate mechanical vibration itself.

Soft Radius / Simple Radius

Supports calculations associated with circular paths and radii.

Correct Hole Drilling

Guides the operator to specified hole coordinates for accurate drilling.

Distance Measurement with Probe

With a touch probe, records point positions and calculates distances between selected points.

Workpiece Angle Measurement

Determines the angle of a workpiece surface relative to the machine axes from selected measurement points.

Last 5 Digits ON/OFF

Allows the display of selected trailing digits to be enabled or disabled according to the required readability.

Tool Diameter Compensation

Accounts for the radius between tool centre and cutting edge when determining machining position.

Hole Grating Function

Generates regular hole-pattern coordinates arranged in rows and columns.

Conical Body Angle Measurement

Calculates angles associated with conical surfaces.

Menu Location Storage

Stores menu/settings locations for faster access to frequently used configurations.

Axis Collection

Allows values from different axes to be combined in calculations.

Machine Reference Point

Provides a defined reference position for machine-coordinate operations.

Team Storage and Selection

Stores and selects sets of working settings.

Tool Memory – 1,000 Pieces

Stores information for up to 1000 tools.

Coordinate Memory – 1,000 Pieces

Stores up to 1000 coordinate records.

Progress Limit

Provides a limit/progress-related machining function as supported by the DRO system.

Axis Value Freeze

Freezes the displayed axis value for a selected operation.

Add / Remove Axis Value

Allows an axis value to be added to or removed from a calculation.

Undo Function

Allows the operator to undo a previous operation.

Sleep Mode

Reduces system activity during periods of non-use.

EDM

Provides an EDM-related operating function.

9.3 Operating and Setup Functions

- Language selection
- Resolution setup
- Measurement-direction setup
- Machine-tool selection
- Compensation-method selection
- Fourth-axis type selection
- Buzzer enable/disable
- Measurement-system change
- Operating-mode selection
- Axis zeroing
- Manual axis-value setting
- Center finding
- Simple axis calculations
- Sleep mode
- Ruler zero point

- Mechanical zero point
- Datum-point memory
- Bolt-hole circle and angular division
- Bolt-hole line
- Smooth radius
- Simple radius
- Linear hole patterns
- Rectangular pocket
- Inclined Z-axis machining

10. Magnetic Linear Measurement System

The linear measurement system associated with the reference DRO uses magnetic measurement components. The system consists of a magnetic read head, magnetic tape, and protective profile.

Component	Specification
Magnetic read head	MLS110
Magnetic tape	B5
Resolution	5 μ m
Output/interface	TTL / 5 VDC / RS-422 line-driver
Signals	A/A, B/B, Z/Z
Cable	3 m shielded spiral cable
Protection	Protective profile

The magnetic scale concept is suitable for machine-tool applications because the sensing system can provide digital displacement information while remaining mechanically independent of the machine's original feed mechanism. Correct alignment between the magnetic tape and read head is essential for reliable measurement.

11. Market Comparison of Position Measurement Systems

Several commercially available DRO systems were reviewed as potential alternatives or references. The original report compares the following products:

System	Manufacturer / Product
A-Tek ADR10-3	ATEK Sensor Technologies
SINO SDS6-3V	SINO
Easson ES-14B	Easson
Acu-Rite DRO203	Acu-Rite
Fagor Innova 30i M	Fagor
Mitutoyo KA-200	Mitutoyo
HEIDENHAIN ND 7013	HEIDENHAIN

The selection should consider axis count, encoder compatibility, resolution, error compensation, environmental robustness, installation requirements, availability, serviceability, and total cost. Prices presented in the original report are market-dependent and should be re-verified before procurement.

12. Alternative Optimization Solutions

12.1 Edge Finder

An edge finder can be used to locate the edge of a workpiece and establish a machining reference. It is a relatively simple solution for specific positioning tasks, but unlike a three-axis DRO it does not continuously display the machine-axis position.

12.2 Touch Probe

A touch probe detects contact between a measurement tip and a workpiece surface or edge and can send a signal to the DRO. It can improve the speed and repeatability of centre finding, edge detection, distance measurement, and angular measurement. However, it is a supplementary measurement tool rather than a complete replacement for a three-axis DRO.

13. CAD Renderings and Installation Design

The CAD work forms a major part of the engineering contribution of the internship. The three-dimensional assembly represents the milling machine and the proposed DRO-related components. The design includes the display unit, magnetic linear measurement components, read heads, mounting elements, and the installation concept for the X, Y, and Z axes.

- Overall milling-machine assembly
- YZ plane / side installation arrangement
- XY plane / table installation arrangement
- XZ plane / vertical-axis installation arrangement
- Y-axis linear encoder
- Z-axis linear encoder
- X-axis linear encoder
- Digital position display
- Mounting brackets and associated mechanical components
- Conceptual cable routing and protection

The installation design is checked for mechanical interference, axis travel, alignment, access for maintenance, and exposure to chips, oil, and vibration. The SolidWorks assembly and drawings provide a basis for fabrication and installation planning.

14. Finite Element Analysis of a Perforated Scrubber Filter

This section is an additional engineering analysis performed during the internship and is separate from the main DRO project. A perforated cylindrical filter used in a scrubber system was evaluated using finite-element analysis.

14.1 Filter Specifications

Parameter	Value
Geometry	Perforated cylindrical filter
Diameter	100 mm
Height	300 mm
Sheet thickness	0.8 mm
Material	Galvanized Carbon Steel

Working pressure	7 bar
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14.2 Analysis Objective

The objective was to investigate the stress level in the perforated filter under the specified pressure condition and to compare two perforation patterns.

14.3 Cases

Case	Hole diameter	Hole pitch	Maximum Von Mises stress
Case 1	3 mm	5 mm	52 MPa
Case 2	5 mm	8 mm	72 MPa

The maximum Von Mises stress increased from 52 MPa to 72 MPa, corresponding to an increase of approximately 38.5%. The result indicates that the perforation geometry has a significant effect on the structural stress distribution of the filter.

14.4 Analysis Workflow

- Geometric modeling of the perforated cylindrical filter
- Definition of material properties
- Application of the working-pressure load
- Definition of boundary conditions
- Generation of the finite-element mesh
- Evaluation using the Von Mises stress criterion
- Comparison and interpretation of the two perforation cases

The original report contains the corresponding geometry, setup, mesh, stress plots, and result images.

15. Conclusion

The internship project provided practical experience in industrial machine investigation, mechanical design, CAD modeling, engineering drawings, technical-document interpretation, and engineering analysis. The main project investigated the improvement of position measurement on an Arsenal FU 321M universal milling machine through a three-axis Digital Readout system.

The study showed that a DRO can provide a more direct and convenient representation of axis position than mechanical graduation reading. The proposed system combines a digital display with linear measurement components on the X, Y, and Z axes. The work also addressed the practical engineering requirements of installation, including alignment, interference, axis travel, cable routing, protection, and maintenance access.

The reference machine and its ATEK ADR10-3 DRO provided a useful basis for understanding system architecture and available functions. Magnetic linear measurement components such as the MLS110 read head and B5 magnetic tape were also examined. Alternative tools such as edge finders and touch probes were considered as complementary solutions rather than direct substitutes for a three-axis DRO.

The CAD development and documentation constitute the main deliverables of the project. The later stages of fabrication, physical installation, calibration, and before/after experimental validation depend on project progress and available time and are therefore not presented here as completed experimental results unless explicitly documented.

In addition to the main DRO study, the internship included a separate finite-element investigation of a perforated scrubber filter. The FEA results demonstrated the effect of perforation geometry on Von Mises stress and provided additional experience in engineering simulation and interpretation.

Appendix A – Key Technical Data

Item	Value
Target machine	Arsenal FU 321M
Target axes	X / Y / Z
Target table	1350 × 320 mm
Target machine mass	2780 kg
Target DRO	ATEK ADR10-3
DRO axis count	3
Linear measurement	Magnetic scale
Read head	MLS110
Magnetic tape	B5
Measurement resolution	5 µm
Signal interface	TTL / 5 VDC / RS-422
DRO supply	85–265 VAC
DRO dimensions	202 × 320 × 84 mm
Coordinate memory	1000
Additional FEA material	Galvanized Carbon Steel
Filter working pressure	7 bar

Appendix B – Main Engineering Skills Applied

- Mechanical Design
- SolidWorks 3D CAD Modeling
- Assembly Design
- Engineering Drawings
- Machine-Tool Study
- Digital Readout (DRO) Systems
- Linear and Magnetic Measurement
- Technical Datasheet Analysis
- Manufacturing and Installation Planning
- Engineering Documentation
- Finite Element Analysis (FEA)
- Stress Analysis and Engineering Interpretation

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